COMPASSION & HAPPINESS

ABSTRACT

If you want others to be happy, practice compassion. If you want to be happy, practice compassion. – Dalai Lama

Many You

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1. INTRODUCTION

In an era increasingly dominated by individualism and self-interest, this data exploration project sets out to explore a fundamental social question: *Are highly compassionate societies happier than those that prioritize self-gain?* Specifically, the project defines a compassionate society as one that upholds empathy, kindness, tolerance, and ethical responsibility within its social, economic, and political systems.

The project is driven by personal observation that as technological progress increases and material prosperity more abundant, social connection, empathy, and kindness often appear to be diminishing which are correlated with the rise in individualism (Santos et al., 2017). For some, one may go to great length such as commiting frauds like crypto scams in order to get ahead (Blue et al., 2024). This trend raises concerns about whether societies that prioritize personal success over collective well-being may be compromising overall happiness.

This project is deeply rooted in a personal belief that compassion plays an essential role in human well-being. The simple act of receiving kindness during difficult times can have a profound emotional impact. Thus, uncovering empirical evidence of a link between compassion and happiness could inspire individuals, communities, and policymakers to foster more compassionate cultures. To guide this exploration, three main questions frame the investigation:

1. **Are societies with higher levels of compassion happier than those with lower levels?**
2. **Do people living in societies with strong social safety nets feel happier?**
3. **What is the correlation between economic inequality and compassion?**

2. DATA SOURCE

To answer these questions, multiple high-quality datasets have been selected:

Dataset	Decription	Purpose
World Value Survey	A large-scale international survey collecting personal opinions and social values from 96,221 respondents across 66 countries between 2018 and 2023.	Estimate compassion levels based on values, social trust, and norms
Source URL: https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp		
OECD Social Expenditure	Official dataset by OECD tracking the percentage of GDP spent on social welfare across 38 OECD member countries from 1980 to 2022.	Measure strength of social safety nets via % of GDP spent on welfare
Source URL: https://www.oecd.org/en/data/datasets/social-expenditure-database-socx.html		
World Happiness Report	An annual report published by the Wellbeing Research Centre at the University of Oxford, ranking 146 countries	Provide national happiness scores for comparison

	based on their average happiness scores for 2022.	
Source URL: https://worldhappiness.report/ed/2022/		
World Bank Gini Coefficient	A global dataset compiled by the World Bank, providing the Gini coefficient (a standard measure of income inequality) for 183 countries between 1960 and 2023.	Measure economic inequality within each country
Source URL: https://data.worldbank.org/indicator/SI.POV.GINI		
World Bank GDP per Capita	A global dataset compiled by the World Bank, providing the GDP capita (a standard measure of population prosperity) for many countries between 1960 and 2023.	Measure the level of individual prosperity within each country
Source URL: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD		

3. DATA PREPARATION

3.1. Preparation Steps

To prepare for our data exploration, we shall combine all of the features that are essential for our analysis into a single main tabular dataset. We would like to work on country level data, so the primary key for the data will be the country code or country name based on ISO 3166-1 alpha 3. One row will represent one country. Ideally, we want the data to be of year 2022 to align with the World Value Survey data, but not every country has information for that year. Thus, we will filter for the year 2022 data, but if it is not available, we will use the most recent data before 2022.

For preparing datasets into a suitable state for our data exploration, we primarily use **Python** to manipulate the dataset, standardize feature naming and input, remove unnecessary features, engineer features, visualize outliers and missing values, join all of the necessary features into a single large dataset for ease of exploration, etcs through the use of **Pandas** library.

The preparation of all datasets shall follow similar steps methods:

1. Read the dataset metadata to understand the nature of each feature in the dataset.
2. Check the formatting of each dataset file (CSV, EXCEL) for whether the dataset is in a proper tabular format by opening the file using tools such as excel.
3. Use appropriate Python functions (such as Pandas) to load and parse the data through functions such as `read_csv()` and `read_excel()` into a Pandas DataFrame object.
4. Format the data into a tabular format if not already
5. Strip the leading and trailing whitespaces within each column names and their inputs to properly spot duplicates that are differentiated through extra white spaces. Space between words in the column name shall be separated by '_' as a standard.
6. Identify various structures of the dataset such as the number of rows and columns, name and data type of each feature, sample inputs of each row, number of missing values and

duplicates, outliers, unique values within a categorical feature by using Pandas function such as: info(), head(), tail(), describe(), isnull(), unique().

7. Clean the datasets such as column names and inputs (categorical) standardization, uncessary columns and duplicated rows removal, data type correction, missing value and outlier handling, long to wide format conversion.
8. Process the datasets such as categorical feature encoding, logarithmic transformation if necessary, year filtering, etcs.

After these steps, the processed datasets shall be joined together into a singular dataset that contains all of the essential information in a wide format.

3.2. Preparation Process

The features that we are interested in extracting for each country are as follows:

country – name of the country

country_code - ISO 3166-1 alpha 3 shorthand for the country

gini – the gini value of a particular country, ranging from 0 to 1.

gdp_per_capita – the gdp per capita vallue of a particular country in $

happiness_score – the happiness score of a country, range from 0 to 10.

percent_gdp_social_spending – net spending on social welfare % of GDP value

compassion_score – calculated compassion level within a country.

The naming of these variables are different across the datasets. They shall be renamed accordingly using the `rename()` Pandas function.

The datasets shall be incrementally processed and merged together from one to another using either **country** or **country_code** as the join key. This flow helps us keep track of what country or year data is available in one dataset and not the other. Thus, we shall start with the least complicated ones first.

Note: Not all transformation steps are described to save space especially repetitive methods.

A. World Bank Gini Coefficient

From this dataset, we shall extract **country, country_code, gini.**

1. Structure of Dataset

The dataset is a 170x3 tabular dataset. It is in a proper csv format, so there is no additional step taken to format the dataset into a proper tabular format and appropriate data types. Each row represents the gini value of a country in a year in long format. There is no duplicate in the dataset, but there are some NULL values in the **country_code**. There is no outlier within the distribution of the gini value. When we check the unique values in the **country** column, we find that there are some regional data within a country as well such as rural and urban data, and it is these observations whose **country_code** feature is empty.

2. Cleaning and Processing

Filter 1: Filter for the year 2022 data or the most recent available year data of a country before 2022 since the World Value Survey data is up to year 2022 since gini value does not change much in mid to short term for any given country unless there is a major societal shift.

Error 1: Missing values in the country_code as described above. **Solution:** We want to use only country wide data since we want to compare the whole country. For country which only has urban or rural data, use the urban data since the Gini index there is the highest within the country and urban area has a large influence on perception within any country.

B. World Bank GDP per capita

From this dataset, we shall extract **gdp_per_capita.**

1. Structure of Dataset

The dataset is a 266x69 tabular dataset. The original csv file is not in a proper tabular format as there are no useful information in the first 4 rows. Thus, the file was parsed using `read_csv()` by skiping the first 4 rows. Each row in the dataset represents a country and has **country** and **country_code** information, and their **gdp_per_capita** information is displayed in a wide format within year columns such as 2020, 2021. There are some NULL values in some year columns, and there is no duplicated rows. There are more unique values for for the country category than there are countries in the world because of overall regional data such as "Asia". There is no outlier in the gdp per capita data.

2. Cleaning and Processing

Filter 1: Select a subset columns that include the last 4 years of data before 2022 in case of missing value in 2022.

Format 1: Convert from Wide to Long format with "country" and "country_code" as the id variables using `melt()` function.

Filter 2: Filter for year 2022 data of each country, or the latest year before 2022 if 2022 is not available. Consequently, this also ensures that the year column will not have any NULL value.

Filter 3 and Merge: Filter out regional data to keep only country level data by performing inner join between the GDP and GINI data with 'country_code' as the join key. What is left will just be the individual country data.

C. World Happiness Report

From this dataset, we extract **happiness_score.**

1. Structure of Dataset

Different from the previous two datasets, this dataset is in an excel spreadsheet format with file extension ".xls". There is also an image within the excel spreadsheet. Fortunately, Pandas can still easily read this type of dataset using `read_excel` function with "xlrd" library installed. What we get after is a proper 147x12 tabular dataset. Each row represent a country level data. The number of

countries within this dataset is less than the previous two datasets. There are missing values in all columns in the last row, and there is no duplicate.

2. Cleaning and Processing

Error 1: Drop last row since it has no values using ` drop().

Format 1: Some country names contain "*" within it such as "Niger*". There is no description about what this refer to on the source website, but we decide to just clean and remove "*" for country name consistency.

Merge: We merge this dataset with the previous dataset, using "country" as the inner join key. But since they do not have the same number of countries, we have to check what are the differences. By using set comparison, we can see that some countries exist in both dataset, but their naming are different such as 'Ivory Coast' and 'Cote d'Ivoire' since the latter is the French name for the the former. Thus, the solution is to map them appropriately before merging the datasets.

D. OECD Social Expenditure

From this dataset, we extract **percent_gdp_social_spending.**

1. Structure of Dataset

The dataset is a 217x34 tabular data. It is in a proper csv format, so there is no additional step taken to format the dataset into a proper tabular format and appropriate data types. We check for the unique values in the country name column, and found that most of the names are consistent with our earlier dataset except 'Turkiye'. There are some null values in columns that are not of our concern, there is no duplicate, and there is no outlier in the net social spending % gdp.

2. Cleaning and Processing

Error 1: Rename 'Türkiye' input in the country column to 'Turkey' using replac() function to ensure consistency across datasets.

Merge: Since there are only a limited number of country data in this dataset (36 countries) compared to over 100 countries in other datasets, we shall perform a left join using 'country_code' as the join key.

E. World Value Survey

From this dataset, we want to calculate the compassion score for each country which is one of the main target variable for our analysis. We shall calculate the score by applying factor analysis algorithm on the interviewee's responds to questions about values, social trust, and norms. Since there are many questions in the survey, we will select a subset of question that are deemed to be related to compassion for simplifying the calculation of the compassion score. There are 15 questions we have selected, and here are some examples and their perceived importance:

Q108 - Government vs individual's responsibility

Evaluates societal perspectives on collective welfare vs. self-interest, reflecting societal compassion towards community support systems.

Q241 - Democracy characteristic: Governments tax the rich and subsidize the poor

Examines societal views on redistribution policies, indicative of compassion toward economically weaker individuals.

1. Structure of Dataset

The dataset is in a proper csv format, but it is large with 97220 rows and 613 columns because of the many respondents and richness of value related questions there were asked. Thus, we shall parse the dataset with only the features that we need. Each row represents an interviewee. The features we need are columns 'B_COUNTRY_ALPHA' which represents the country code, 'D_INTERVIEW' which represents the interview, and the answers to the 15 survey questions we mentioned above. There is no NULL or duplicate value in the dataset.

The interviewee's answer to each question is recorded in ordinal value such as below:

Q108	**Government´s vs individual´s responsibility** *Now I'd like you to tell me your views on various issues. How would you place your views on this scale? 1 means you agree completely with the statement on the left; 10 means you agree completely with the statement on the right; and if your views fall somewhere in between, you can choose any number in between.* 1.- The government should take more responsibility to ensure that everyone is provided for 2.- 2 3.- 3 4.- 4 5.- 5 6.- 6 7.- 7 8.- 8 9.- 9 10.- People should take more responsibility to provide for themselves -1-.- Don´t know

However, the ordinal scale is not uniform across the questions. For examples, some range from -4 to 4, and some range from -4 to 10 depending on the nature of the questions. In all instances, negative values indicate that there is no response given.

2. Cleaning and Processing

Filter 1: Any respondent who does not answer any one of the question subset we have is filtered out to ensure that the weight of the question for our compassion score calculation is not affected. We can do this by using using the greater or equal function `ge()` on the selected columns.

Format 1: Some answers have reverse meaning in their scales. For some, the lower the value, the higher indication of compassion level such as for Q108 above which is counter intuitive. Thus, the scales are reversed by adding up the max and min values in that column and minus the input value.

Format 2: Factor Analysis algorithm is sensitive to the scaling. Since the ordinal scaling of each question answer is not uniform, we have to apply Min-Max scaling which can be done with the MinMaxScaler() from the `**sklearn**` library, so that the range of value is between 0 and 1.

Feature Engineer 1: After the previous steps, we can now apply the factor analysis algorithm from `**factor_analyzer**` library to estimate the compassion score of each interviewee. Then, the compassion score is averaged and grouped by the country that the respondent is from.

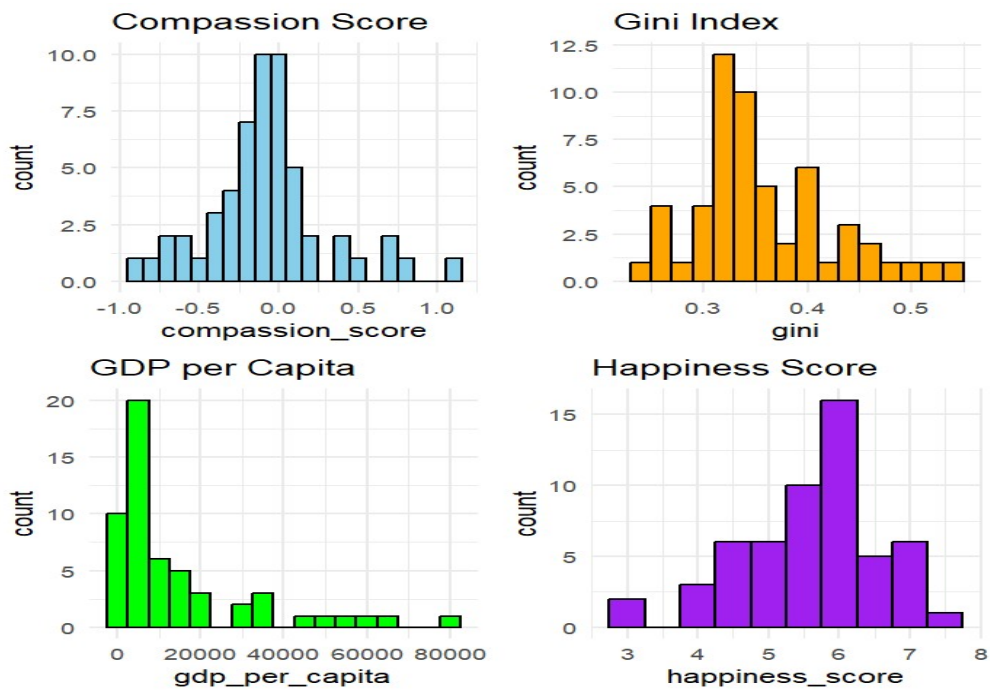
Merge: The generated compassion score for each country shall be inner joined with the data with the previous dataset using country_code as the join key. Since the compassion score dataset does not have every country that is in the previous dataset, we only have 55 countries left afterwards.

4. DATA EXPLORATION AND CONCLUSION

For our data exploration process, we shall use visualizations in R to help understand the underlying data and answer our three questions.

4.1. Understand the Underlyng Structure of the Dataset

Visualizing the distribution of variables is a crucial step in data exploration because it reveals the underlying shape, spread, and patterns that are intuitive to human.



Based on the graphs, we can interprete as follows:

Compassion Score: Roughly normally distributed, centered around 0. Most countries cluster near the center, so most countries have moderate level of compassion.

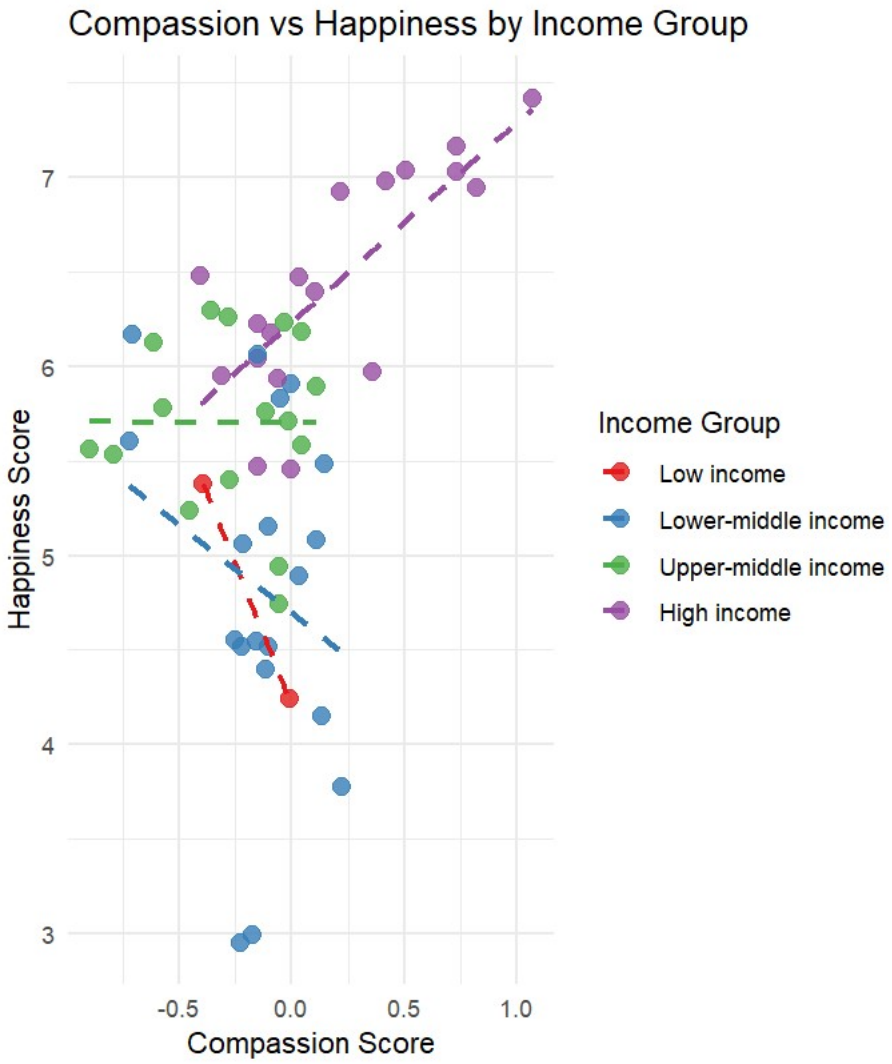
Gini Index: Slightly right-skewed, unimodal. Most countries fall between 0.30–0.40, indicating moderate inequality. A few countries exceed 0.45, which reflects high income inequality.

GDP per capita: Heavily right-skewed. Most countries are on the lower-income end (<$30K). A few wealthy outliers (e.g. Luxembourg, Switzerland) stretch the tail. It might be interesting to analyze the compassion level and happiness score based on GDP per capita grouping.

Happiness Score: Slight left-skew, but mostly symmetric. Most countries score between 5.5 and 7.0, indicating generally high happiness. Only a few countries show very low happiness (<5)

B. Answering Questions

1. Are societies with higher levels of compassion happier than those with lower levels?



The scatter plot visualizes the relationship between compassion score and happiness score with color showing the income group category the country data belongs to. The income group is allocated based on the classification of World Bank (Metreau et al., 2024). Based on the graph, we interprete as follows:

A. High Income: Higher compassion scores are consistently associated with higher happiness scores. The positive slope is steep and distinct, suggesting compassion might play an important role in happiness for wealthy countries.

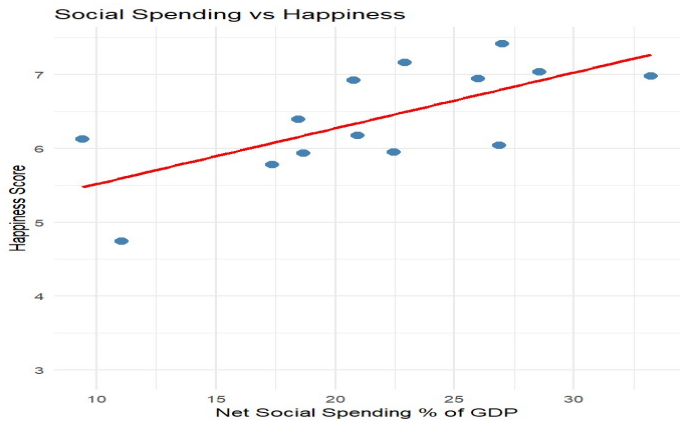
B. Upper-middle Income: The relationship appears relatively flat, indicating that compassion scores might not strongly predict happiness in these countries, although slightly upward trend indicates mild positive association. Other economic or social factors could be playing a more dominant role in influencing happiness

C. Lower-middle Income: Surprisingly, compassion shows a negative relationship with happiness here that is higher compassion appears associated with slightly lower happiness. Compassion might reflect different realities in lower-middle income contexts. For example, high compassion might reflect necessity and hardship, rather than voluntary empathy. It could also indicate structural factors overshadowing compassionate culture in influencing happiness (e.g., economic instability, insecurity).

D. Low income: Again, a noticeable negative relationship, although data points are limited. Similar to lower-middle income group, compassion might correlate negatively with happiness possibly due to difficult socioeconomic realities dominating over compassionate cultural traits.

Conclusion: The relationship between compassion and happiness varies by a country's income level. For high-income nations, a strong positive relationship suggests compassion significantly enhances happiness, likely due to established social structures and security. In contrast, this relationship is weaker or even negative in lower-income contexts, suggesting that economic or social hardships might overshadow the benefits of compassion on well-being.

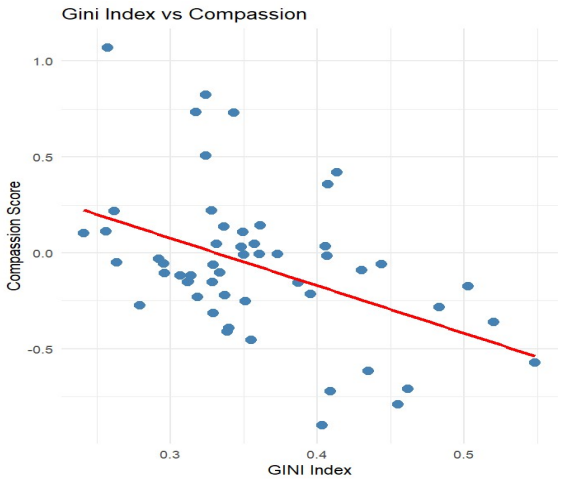
2. Do people living in societies with strong social safety nets feel happier?



The graph visualizes the linear relationship betwween Social Spending and Happiness. There is a strong positive correlation such that as social spending increases, happiness scores tend to rise. This suggests a direct link between the strength of a country's social safety net and its population's well-being. Stronger government investment in public support services (like healthcare, pensions, unemployment aid) appears to support a higher quality of life. This plot only includes countries with available data on social spending, so results might be biased toward OECD or high-income countries.

Conclusion: The graph provides compelling evidence that countries with higher social spending tend to be happier, supporting the idea that robust social safety nets contribute to well-being. This insight aligns with the idea that compassionate societies, backed by institutional support, foster higher happiness levels. However, while social spending and happiness are linked, other factors (like governance quality, culture, economic stability) may also contribute.

3. What is the correlation between economic inequality and compassion?



The graph visualizes the linear relationship between inequality (GINI index) and our estimated compassion score. The red trendline slopes downward, indicating a moderate to strong negative correlation. This graph supports the idea that more equal societies tend to be more compassionate. Lower inequality may foster greater trust, social cohesion, and prosocial values — all of which reflect in higher compassion scores. In contrast, more unequal societies may experience more division and less collective concern, suppressing compassionate norms.

Conclusion: A notable negative relationship exists between economic inequality and compassion that is more equal societies typically demonstrate higher compassion levels. High inequality seems to erode social trust and empathy, reducing compassionate behavior and attitudes. However, while the correlation is clear, causation is not implied. We cannot say inequality causes low compassion or vice versa.

5. REFLECTION

This project has taught me to think broadly about the different dimensions that I may ask questions related to the topic, especially something as abstract as compassion and happiness in my case. I have to be able to analyze their relationships across different domains such as the level of income, inequality, etcs. However, this caused me to have worked with many datasets which make my data wrangling process bothersome.

In addition to that, formulating the compassionate score took a bit of work from me because there are so many value related questions (around 300 questions) I can choose from the World Value Survey dataset. In hindsight, I should not have selected such an abstract topic like this because of the limited time I have. Nevertheless, it was exciting to see that the questions I picked led to the generation of the compassion score that is correlated meaningfully with other variables in the explorations that I did.

Lastly, the availability of data for some countries are limited. For example, at the end we only have 55 countries data to work with, limited by the World Value Survey. Such limitation forced me to become creative such as using older data if necessary as long as I do not expect them to have changed much.

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